The *S.T.A.R. Program
(Symbolic-Regressing Hyperparamater-Tuned Astronomical Regression)

A Comprehensive Study: BSD Cosmology & Star Formation Rate Prediction

Abstract

This study introduces a novel theoretical framework that bridges number theory and astrophysics.

The *S.T.A.R. Program for a Symbolic-Regressing, Hyperparamater-Tuned Astronomical Regression machine learning model.

Inspired by the Birch and Swinnerton-Dyer (BSD) conjecture, to construct a cosmological L-function to model and predict the *Star Formation Rate (SFR) in galaxies.

The underlying hypothesis is that the mathematical properties of this L-function, particularly its behavior near s = 1, reflect fundamental drivers of SFR.

The model is developed and tested using galaxy data from the Sloan Digital Sky Survey (SDSS), demonstrating promising results and setting the stage for future research.

1. Introduction

1.1 Background

The BSD conjecture, a central unsolved problem in number theory, posits a deep connection between elliptic curves and L-functions.

It states that the rank of an elliptic curve (the number of independent rational points) is equal to the order of the zero of its associated L-function at s = 1.

This conjecture has inspired applications far beyond pure mathematics.

In astrophysics, the star formation rate (SFR) is a crucial parameter for understanding galaxy evolution.

SFR is influenced by factors such as stellar mass, redshift, gas density, metallicity, and environmental effects.

Traditional SFR models range from empirical relations (e.g., the Kennicutt-Schmidt law) to complex simulations (e.g., IllustrisTNG), each with limitations in scope and predictive power.

1.2 Motivation and Objectives

This study aims to develop a mathematically grounded model for SFR prediction based on the BSD conjecture. Our main objectives are:

- To establish a theoretical analogy between elliptic curves and galaxy distributions.

- To construct a cosmological L-function relevant to SFR dynamics.

- To integrate this L-function into a predictive model and assess its performance using observational data.

1.3 Research Questions and Hypotheses

Addressing the following questions:

- Can a cosmological L-function be constructed such that its mathematical properties reflect SFR drivers?

- Does incorporating this L-function improve SFR prediction accuracy compared to traditional models?

Our central hypothesis is that the BSD conjecture provides a viable mathematical framework for understanding and modeling SFR.

2. Theoretical Framework

2.1 The Birch and Swinnerton-Dyer Conjecture: A Brief Overview

Consider an elliptic curve E defined over the rational numbers (Q).

The BSD conjecture relates the arithmetic of E to the analytic behavior of its L-function, L(E, s).

Specifically, the BSD conjecture states:

• The rank of E, denoted rank(E), equals the order of the zero of L(E, s) at s = 1.

• The leading coefficient of the Taylor expansion of L(E, s) around s = 1 encodes additional arithmetic data, providing deeper insights into the curve's properties.

This conjecture suggests that complex distributions of rational points can be analyzed via analytic functions, a concept I extend to cosmology.

2.2 Analogy to Galaxy Distributions

In “BSD Cosmology”, galaxies are treated as points in a multidimensional parameter space, with coordinates such as logarithmic stellar mass (logmass), redshift (z), declination (dec), and Petrosian radius (petrorad_r).

Just as the rank of an elliptic curve reflects the number of independent generators of its rational points, the proposition is that the "rank" of a galaxy distribution reflects the number of independent physical processes driving SFR.
(e.g., gas accretion, stellar feedback).

2.3 Defining the Cosmological L-Function

To operationalize this analogy define a cosmological L-function:

$$L_{cosmo}(s) = \sum a_n n^{(-s)},$$

where:

• n indexes bins in the galaxy parameter space (e.g., discretized by mass and redshift).

• a_n is the number of galaxies in bin n, optionally weighted by physical properties like SFR or density.

The behavior of Lcosmo(s) at s = 1 is hypothesized to indicate the number of independent SFR drivers, mirroring the BSD conjecture.

3. Model Development

3.1 Constructing the L-Function

To operationalize Lcosmo(s):

• Binning: Discretize galaxy properties into a grid. For example, logmass in 0.1 dex intervals and z in 0.01 intervals.

• Coefficients: Set a_n as the galaxy count per bin, weighted by observed SFR to emphasize star-forming regions.

• Analytic Continuation: Approximate Lcosmo(s) numerically, focusing on its value and derivatives at s = 1.

For simplicity, initially assume a rank-1 scenario (one dominant driver), adjusting the model as data suggests higher complexity.

3.2 SFR Prediction Formula

The SFR is modeled as:

SFR = a * logmass + b * z + c * Lcosmo(1) + d * environment + e * petrorad_r,

where:

• logmass and z are primary predictors from cosmological literature.

• Lcosmo(1) introduces the BSD-inspired term.

• environment (derived from spatial coordinates)

and petrorad_r account for additional physics.

• Coefficients a, b, c, d, and e are optimized via regression.

3.3 Implementation Details

Using datasets of galaxy properties to:
(e.g., from the SDSS, MangaHIall etc.)

• Preprocess data: Handle missing values, normalize features.

• Compute Lcosmo(s) over binned data.

• Fit the model using:
• a Random Forest Regressor
• a Gradient Boosting Regressor
• a GPLearn Symbolic Regressor
• an Optuna Optimization Tuner

Chosen for their abilities to capture non-linear interactions, zero in on patterns, and verify results using multiple verification methods.

Further testing will include raw unfiltered data.

4. Implementation and Testing

4.1 Data Sources

The goal of using raw, unfiltered data to improve robustness of the analysis in the context of the *S.T.A.R. Program, the following appropriate VizieR catalogues align with the focus on galaxy properties, star formation rates, and BSD-inspired features.

The program works with galaxy-wide measurements (e.g., log_SFR_Ha, log_Mass_gas, Re_kpc, OH_O3N2_cen) and aims to model star formation rates while incorporating cosmological influences through features like cosmo_rank and L_cosmo_s*.

Using raw data to preserve observational variability and reduce biases from preprocessing, prioritizing catalogues with minimally processed measurements (e.g., fluxes, magnitudes) relevant to galaxies.

With direct links to these catalogues on VizieR, ensuring they can be accessed for analysis.

The program is designed to:
Model log_SFR_Ha (star formation rate derived from Hα emission) for galaxies.

Use features like log_Mass_gas, Re_kpc (effective radius), and metallicity indicators (OH_O3N2_cen).

Incorporate BSD features (cosmo_rank, L_cosmo_s*) that depend on galaxy properties like mass and size.

Improve robustness by using raw data, meaning to preserve observational noise (e.g., measurement errors, systematic effects) rather than relying on preprocessed or derived quantities.

Raw Data in Astronomy: In this context, "raw unfiltered data" refers to measurements closer to direct observations—such as fluxes, magnitudes, or spectral line intensities—before they are heavily processed into derived quantities like SFRs or metallicities.

While no catalogue provides truly raw telescope data (e.g., photon counts), some VizieR catalogues offer minimally processed data (calibrated but not aggregated or cleaned), which aligns with the goal.

Requirements for Catalogues:

Using data from these coordinates:
134.6125 + 57.2325
150 arcmin around
Rad. 5° arcsec

Object Type: Must focus on galaxies, not stellar systems like cataclysmic variables (e.g., III/286 was unsuitable).

Data Type: Should provide raw or minimally processed measurements (e.g., emission line fluxes, photometry) relevant to star formation, mass, or metallicity.

Cross-Matching: Must include coordinates (RA, Dec) to merge with the dataset (objra_y, objdec).

Robustness: Should preserve observational variability (e.g., include errors, avoid heavy outlier removal).

VizieR catalogues that meet the criteria, with explanations of their relevance, the type of raw data they provide, and direct links to access them.

Focusing on SDSS-based catalogues (and complementary surveys like 2MASS and GAMA) which provide raw measurements for galaxies.

1. SDSS DR16 Spectroscopic Data (V/154)

Why It's Suitable:

Contains spectroscopic measurements for galaxies, including raw emission line fluxes (e.g., Hα, Hβ, [OIII], [NII]), which are directly relevant to the target (log_SFR_Ha) and features (e.g., OH_O3N2_cen).

Provides minimally processed data: fluxes are calibrated (e.g., corrected for instrumental effects) but not aggregated or heavily cleaned, preserving observational noise.

Includes measurement errors (e.g., e_flux_6563), which can enhance robustness by exposing the model to uncertainty.

Matches the SDSS origin dataset, increasing the likelihood of successful cross-matching.

Key Columns:

RA, Dec (for cross-matching with objra_y, objdec).

specz (spectroscopic redshift, useful for distance calculations).

flux_6563 (Hα flux, can be used to recompute log_SFR_Ha).

flux_4861 (Hβ flux, for extinction correction via Hα/Hβ ratio).

flux_5007 ([OIII] 5007 flux, for metallicity via O3N2).

flux_6584 ([NII] 6584 flux, for metallicity).

e_flux_6563 (error in Hα flux, captures observational uncertainty).

How It Improves Robustness:

Raw fluxes include measurement errors and systematic effects (e.g., extinction, aperture corrections), making the model more resilient to real-world data imperfections.

Recomputing log_SFR_Ha and metallicity from fluxes introduces variability not present in preprocessed values.

Link to Catalogue:

Access V/154 on VizieR: http://vizier.cds.unistra.fr/viz-bin/VizieR?-source=V/154

Query by coordinates to match the dataset or download the full table (full table chosen)

2. SDSS DR12 Photometric Catalogue (V/147)
Why It's Suitable:

Provides photometric measurements (magnitudes in u, g, r, i, z bands) for millions of galaxies from SDSS DR12.

Raw Data: Offers PSF, Petrosian, and model magnitudes directly from observations, along with errors, without heavy processing beyond calibration.

Complements spectroscopic data by adding optical photometry, which can be used to derive galaxy colors (e.g., g-r, u-g) as proxies for stellar population age and star formation history.

Key Columns:

RA, Dec.

petroMag_u, petroMag_g, petroMag_r, petroMag_i, petroMag_z (Petrosian magnitudes in SDSS bands).

petroMagErr_u, petroMagErr_g, etc. (magnitude errors).

z (photometric redshift, if spectroscopic redshift is unavailable).

How It Improves Robustness:

Raw magnitudes introduce variability due to dust, observational conditions, and measurement errors, reducing overfitting to cleaned data.

Galaxy colors can capture additional variance in star formation activity, complementing log_SFR_Ha.

Link to Catalogue:

Access V/147 on VizieR: http://vizier.cds.unistra.fr/viz-bin/VizieR?-source=V/147

Query by position or download subsets (PhotoObjAll table chosen)

3. 2MASS Extended Source Catalogue (II/246)

Why It's Suitable:

Provides near-infrared photometry (J, H, Ks bands) for extended sources (mostly galaxies) from the Two Micron All Sky Survey (2MASS).

Raw Data: Includes magnitudes and fluxes with errors, minimally processed beyond calibration.

Near-infrared data is less affected by dust extinction, providing a complementary view of stellar mass (correlates with log_Mass in the dataset).

Key Columns:
RA, Dec.

Jmag, Hmag, Kmag (extrapolated magnitudes in J, H, Ks bands).

e_Jmag, e_Hmag, e_Kmag (magnitude errors).

r_ext (extrapolated radius, a proxy for galaxy size).

How It Improves Robustness:

Adds a different wavelength regime (near-infrared), exposing the model to galaxies with varying dust content.

Magnitude errors capture observational uncertainty, enhancing generalizability.

Link to Catalogue:

Access II/246 on VizieR: http://vizier.cds.unistra.fr/viz-bin/VizieR?-source=II/246

Focusing on the main table (xsc) for extended sources.

4. GAMA DR3 Spectroscopic and Photometric Data (II/356)

Why It's Suitable:

The Galaxy And Mass Assembly (GAMA) survey provides both spectroscopic and multi-wavelength photometric data for galaxies, overlapping with SDSS but extending to deeper magnitudes.

Raw Data: Includes emission line fluxes (e.g., Hα, [NII]) and photometry across 20 bands (UV to far-IR), minimally processed.

Particularly useful for recomputing log_SFR_Ha and adding multi-wavelength features.

Key Columns:

RA, Dec.

Flux_Ha (Hα flux from spectroscopy).

Flux_NII_6584 ([NII] 6584 flux).

u_mag, g_mag, r_mag, etc. (photometry from SDSS and other surveys).

e_Flux_Ha (error in Hα flux).

How It Improves Robustness:

Combines spectroscopy and photometry, providing a comprehensive view of galaxy properties with observational noise intact.

Multi-wavelength data helps capture diverse physical processes (e.g., star formation, dust), improving model generalizability.

Link to Catalogue:

Access II/356 on VizieR: http://vizier.cds.unistra.fr/viz-bin/VizieR?-source=II/356

GamaSpec (spectroscopy) and GamaPhoto (photometry)

4.2 Data Preprocessing

The data preprocessing steps are:

• Cleaning: Impute missing values with the

median for each column.

• Feature Engineering: Create an environmental density proxy from the spatial clustering of dec and right ascension.

• Normalization: Normalize all input features to a range between 0 and 1 to ensure stable regression performance.

4.3 Model Training and Validation

The model is trained and validated using:

• 5-fold cross-validation.

• Mean Squared Error (MSE) between predicted and observed SFR as the evaluation metric.

•Tuning of Random Forest hyperparameters (e.g., tree depth, number of estimators) to optimize performance.

5. Results and Analysis

5.1 Numerical Evaluation of Lcosmo(s)

Numerical evaluation suggests Lcosmo(s) has a first-order zero at s = 1 (rank 1), implying one dominant SFR driver (e.g., stellar mass), with secondary factors refining the prediction. This aligns with cosmological intuition that mass is a primary driver.

5.2 Model Performance

The BSD Cosmology model demonstrates the following performance:

• Average MSE: 0.4266
(logarithmic units, normalized).

• Standard Deviation: 0.0185 across folds, indicating consistency.

• Comparison: An earlier model without Lcosmo(1) or environmental terms yielded MSE = 0.6715, demonstrating improvement with the inclusion of the BSD-inspired term.

5.3 Interpretation

The results suggest:

• The BSD-inspired Lcosmo(1) term reduces reliance on empirical coefficients, grounding the model in a mathematical framework.

• The model accurately captures key SFR drivers, as evidenced by the low MSE and stable cross-validation performance.

6. Conclusion and Implications

6.1 Conclusion

This study introduces "BSD Cosmology," a novel theory merging the BSD conjecture with SFR prediction.

By constructing a cosmological L-function and integrating it into a predictive model, achieving promising results, outperforming baseline approaches.

6.2 Implications for Science and Mathematical Logic

This work has implications for both science and mathematical logic:

• Mathematical Logic: Demonstrates how number-theoretic tools (L-functions, rank) can model physical phenomena, potentially bridging

pure mathematics and applied sciences.
It also suggests a new class of L-functions for astrophysical distributions.

• Scientific Impact: Enhances SFR prediction accuracy, aiding galaxy evolution studies. It also offers a mathematically rigorous alternative to empirical models, testable with observational data (e.g., SDSS, JWST).

7. Future Directions

The model can be refined in several ways:

• L-Function Enhancement: Incorporate metallicity or gas mass into a_n.

• Feature Expansion: Add star formation history or supernovae feedback.

• Model Optimization: Test Gradient Boosting or neural networks.

• Validation: Apply to external datasets (e.g.,GAMA survey) for generalizability.

8. Summary

The BSD Cosmology theory reimagines SFR prediction through a number-theoretic lens, using a cosmological L-function inspired by the BSD conjecture.

The model, tested on galaxy data, achieves strong performance, resolving errors through careful preprocessing and debugging.

Its implications span mathematics and cosmology, offering a framework that could deepen our understanding of the universe while showcasing the power of interdisciplinary approaches.

Future refinements promise even greater precision and insight.

Further studies to follow.

With
PanSTARRS DR1,
SDSS DR12,
SDSS DR16

Coordinates
134.6125 + 57.2375
150 arcmin around
Rad. 60° arcsec

9. References

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